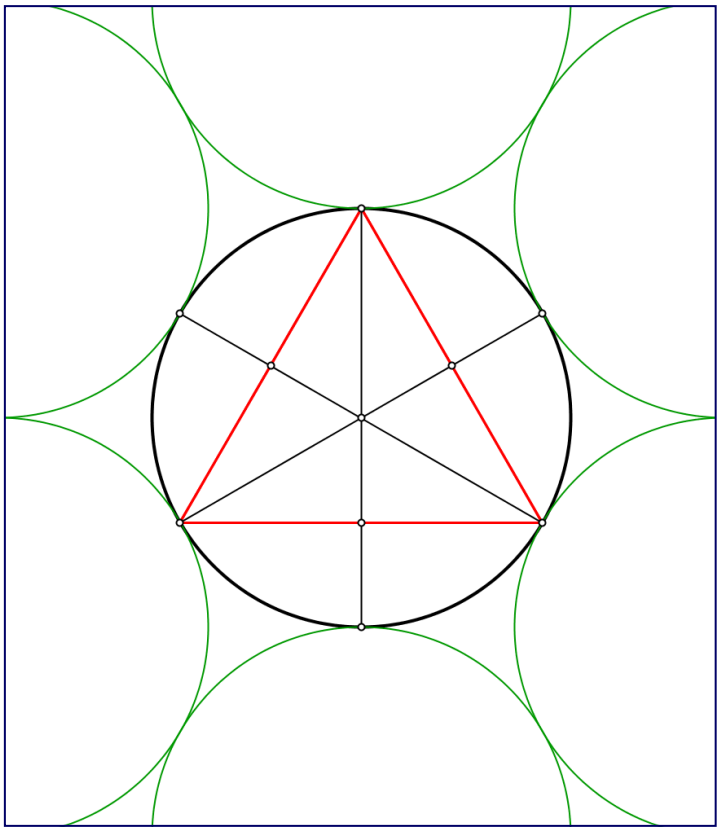


RESUMEN RMA-03
GEOMETRÍA I

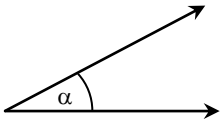
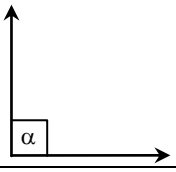
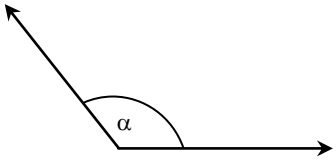
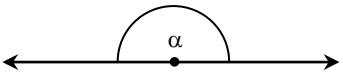
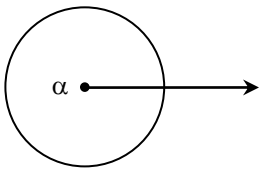
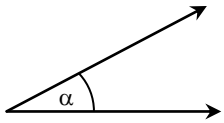
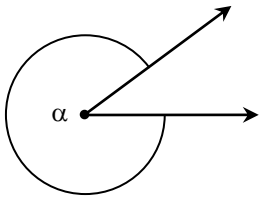
Nombre _____

Curso _____

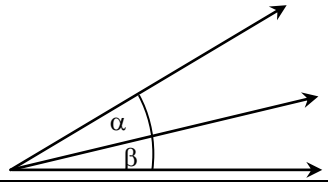
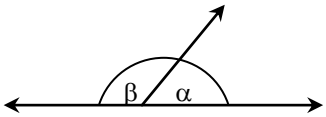
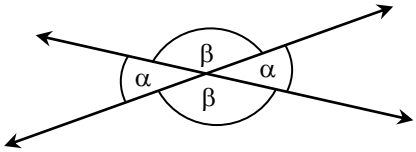
Profesor _____

PREUNIVERSITARIO
PEDRO DE VALDIVIA

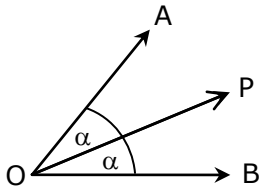
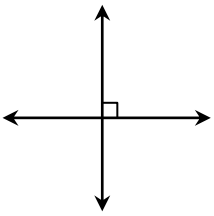
ÁNGULOS

NOMBRE DEL ÁNGULO	MEDIDA	FORMA
Nulo	$\alpha = 0°$	
Agudo	$0° < \alpha < 90°$	
Recto	$\alpha = 90°$	
Obtuso	$90° < \alpha < 180°$	
Extendido	$\alpha = 180°$	
Completo	$\alpha = 360°$	
Convexo	$0° < \alpha < 180°$	
Cóncavo	$180° < \alpha < 360°$	

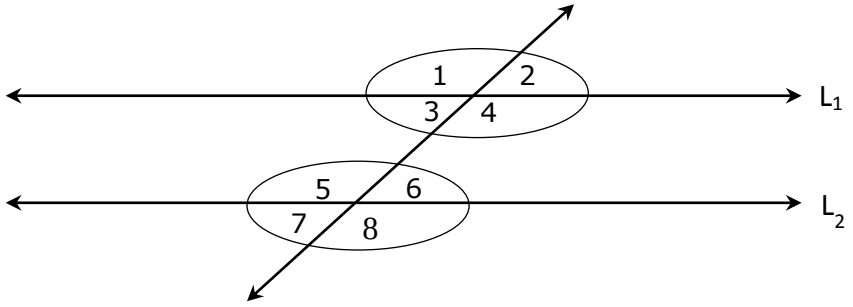
CLASIFICACIÓN DE LOS ÁNGULOS SEGÚN SU POSICIÓN

NOMBRE	IMAGEN	CARACTERÍSTICA
Ángulos Consecutivos		♦ Tienen el vértice y un rayo común que los separa, y no está uno comprendido en el otro.
Ángulos Adyacentes		♦ Tienen el vértice y un rayo en común. Los otros dos rayos están en una recta común. ♦ Son consecutivos. ♦ $\alpha + \beta = 180°$
Ángulos Opuestos por el Vértice		♦ Se forman al intersectar dos rectas y son congruentes.

OBSERVACIONES:

Bisectriz de un Ángulo		♦ $\overrightarrow{OP}$ es bisectriz ♦ $\overrightarrow{OP}$ divide al ángulo AOB en dos ángulos de igual medida.
Rectas Perpendiculares		♦ Las rectas se cortan formando un ángulo de 90°.

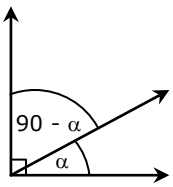
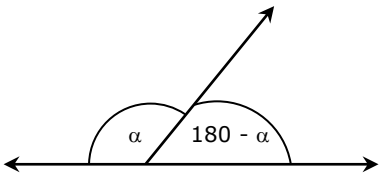
ÁNGULOS FORMADOS POR DOS RECTAS PARALELAS CORTADAS POR UNA TRANSVERSAL



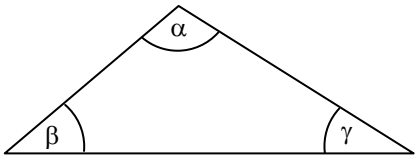
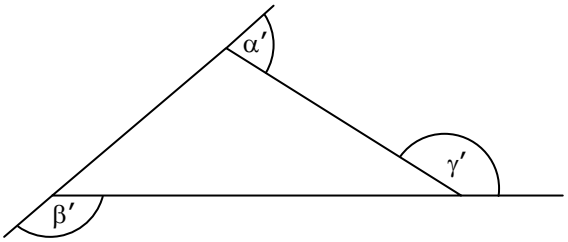
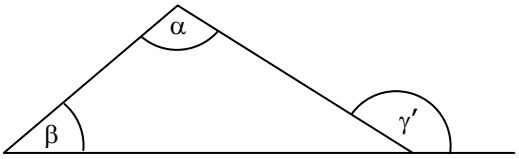
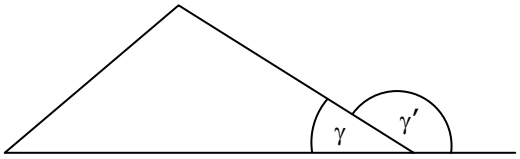
L_1 // L_2

Ángulos Alternos	**Internos:** ∡3 con ∡6 ∡4 con ∡5	**Son congruentes**
	Externos: ∡1 con ∡8 ∡2 con ∡7	
Ángulos Correspondientes	♦ ∡1 con ∡5 ♦ ∡2 con ∡6 ♦ ∡3 con ∡7 ♦ ∡4 con ∡8	**Son congruentes**
Ángulos Colaterales	♦ ∡1 , ∡3 , ∡5 y ∡7 ♦ ∡2 , ∡4 , ∡6 y ∡8	**Son suplementarios**

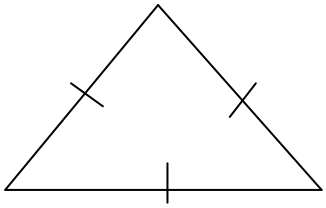
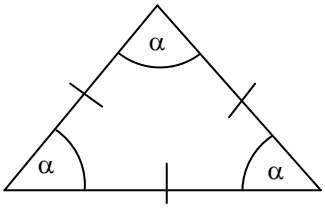
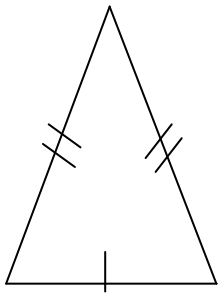
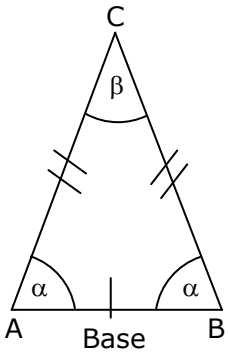
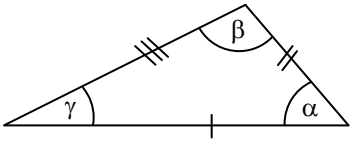
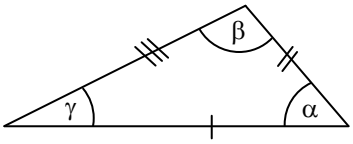
OBSERVACIÓN:
Los Ángulos Alternos y Ángulos Correspondientes son congruentes entre sí.

Ángulos	**Complementarios**	♦ Suman 90° ♦ Complemento de α = 90° - α	
	Suplementarios	♦ Suman 180° ♦ Suplemento de α = 180° - α	

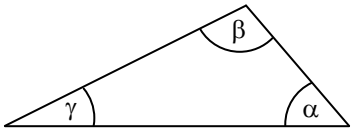
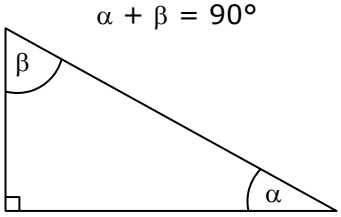
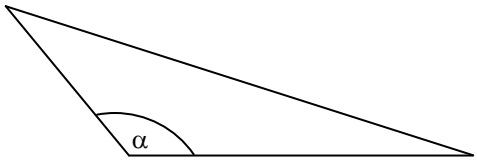
TRIÁNGULOS

TEOREMA	IMAGEN	SE CUMPLE
Suma de los ángulos interiores		$\alpha + \beta + \gamma = 180°$
Suma de los ángulos exteriores (adyacentes a los ángulos interiores)		$\alpha' + \beta' + \gamma' = 360°$
Relación entre dos ángulos interiores y el exterior no adyacente a ellos		$\alpha + \beta = \gamma'$
Observación		$\gamma + \gamma' = 180°$

CLASIFICACIÓN DE TRIÁNGULOS

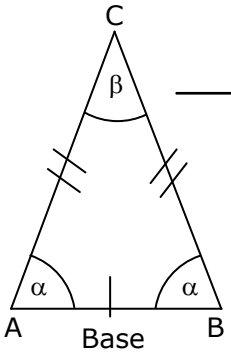
	NOMBRE	CONDICIÓN	REPRESENTACIÓN GRÁFICA	CONSECUENCIAS DE LA CLASIFICACIÓN
Según sus Lados	**Equilátero**	Tres lados congruentes		 $\alpha = 60°$
	Isósceles	Dos lados congruentes		
	Escaleno	Sin lados congruentes		$\alpha \neq \beta \neq \gamma$ 

Observación:
Si un triángulo tiene dos lados de igual medida dicho triángulo es al menos isósceles, pues si el tercer lado es de igual medida que los otros dos, entonces será equilátero.

	NOMBRE	CONDICIÓN	REPRESENTACIÓN GRÁFICA
Según sus Ángulos	**Acutángulo**	Tres ángulos agudos	$0° < \alpha < 90°$ $0° < \beta < 90°$ $0° < \gamma < 90°$ 
	Rectángulo	Un ángulo recto	$\alpha + \beta = 90°$ 
	Obtusángulo	Un ángulo obtuso (α)	 $90° < \alpha < 180°$

OBSERVACIONES:

Triángulo Isósceles:

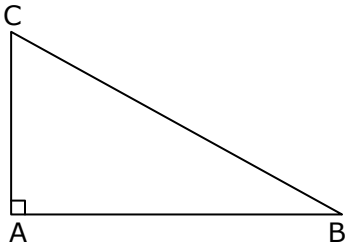


Ángulo del vértice

α se denomina ángulo basal.

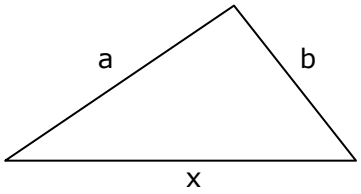
Triángulo rectángulo

- Catetos: $\overline{AB}$ y $\overline{AC}$ (forman el ángulo recto)
- Hipotenusa: $\overline{BC}$



DESIGUALDAD TRIANGULAR

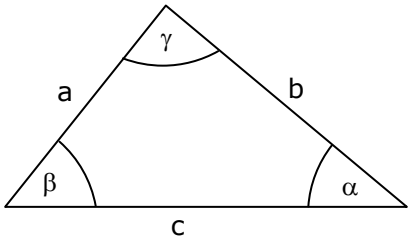
En todo triángulo, el tercer lado siempre mide menos que la suma de los otros dos, y, más que la diferencia positiva de esos mismos dos lados.



$$|a - b| < x < a + b$$

RELACIÓN ENTRE EL LADO DE UN TRIÁNGULO Y EL ÁNGULO QUE SE OPONE A ÉSTE

"A mayor lado, mayor es el ángulo que se opone a dicho lado y viceversa"

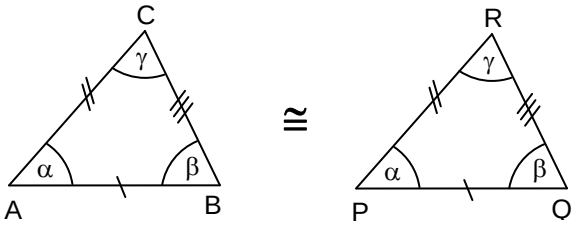


$$a < b < c \Leftrightarrow \alpha < \beta < \gamma$$

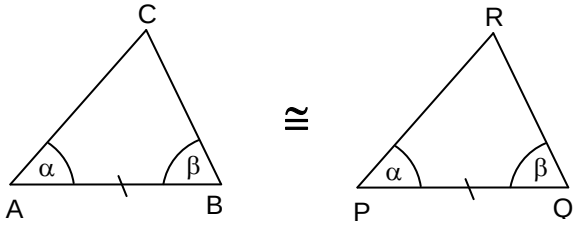
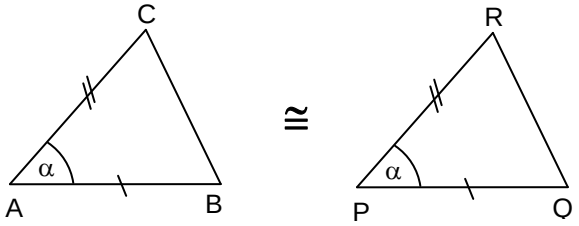
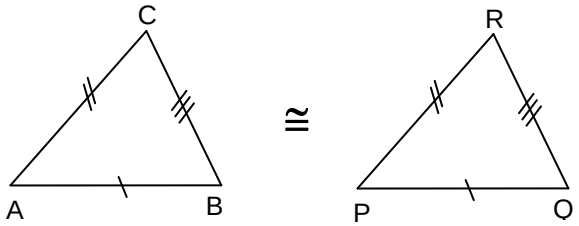
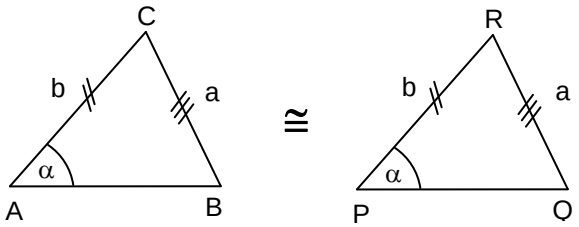
$$a = b \Leftrightarrow \alpha = \beta$$

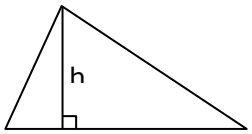
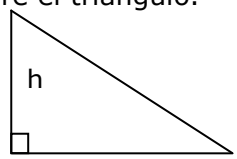
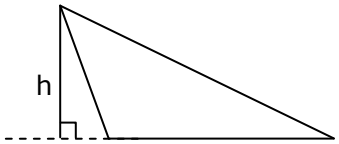
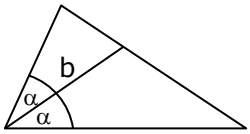
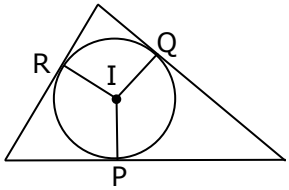
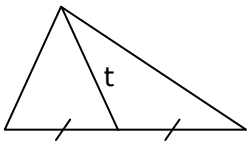
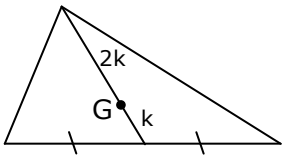
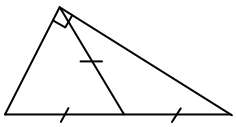
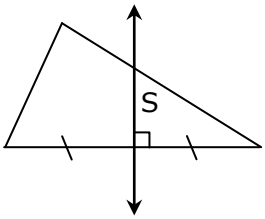
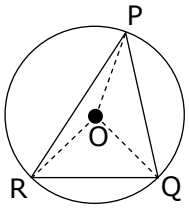
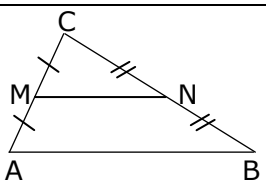
CONGRUENCIA DE TRIÁNGULOS

$\Delta ABC \cong \Delta PQR$

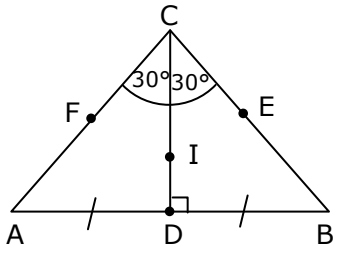
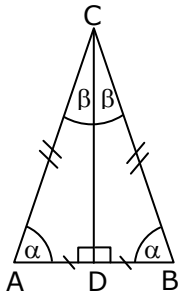


POSTULADOS DE CONGRUENCIA

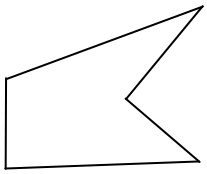
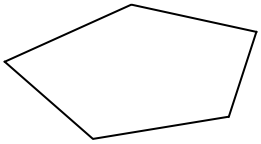
POSTULADO	REPRESENTACIÓN GRÁFICA
ALA	
LAL	
LLL	
LLA> **con a > b**	

ELEMENTOS SECUNDARIOS DEL TRIÁNGULO				
NOMBRE	**REPRESENTACIÓN GRÁFICA**	**PUNTO SINGULAR**	**DEFINICIÓN**	**OBSERVACIONES**
Altura (h)		Ortocentro (H)	Cae desde un vértice y forma un ángulo de 90° con la recta que contiene al lado opuesto.	♦ La altura puede caer sobre el triángulo.  ♦ La altura puede caer fuera del triángulo 
Bisectriz (b)		Incentro (I)	Divide al ángulo en dos ángulos congruentes.	"I" equidista de los lados del triángulo.  IP = IQ = IR = Radio
Transversal de Gravedad (t)		Centro de Gravedad (G)	Segmento que se origina desde un vértice e intersecta en el punto medio del lado opuesto.	El punto "G" divide a la transversal en razón 2:1.  
Simetral (S)		Circuncentro (O)	Recta perpendicular, exactamente en el punto medio, de un lado del triángulo.	"O" equidista de los vértices del triángulo.  OP = OR = OQ
Mediana ($\overline{MN}$)		No hay	Segmento que une los puntos medios de dos lados de un triángulo.	♦ $\overline{MN} \parallel \overline{AB}$ ♦ Mediana = MN = $\frac{AB}{2}$

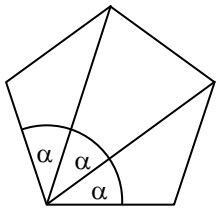
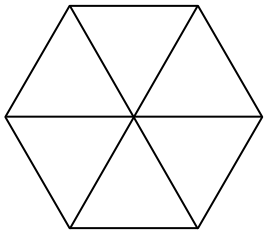
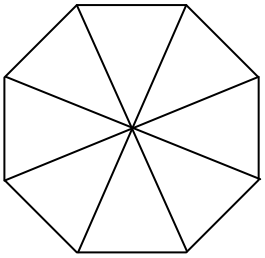
OBSERVACIONES

TRIANGULO EQUILÁTERO	
	♦ Coinciden todos los elementos secundarios. ♦ CD = Altura = Bisectriz = Transversal de Gravedad = Simetral ♦ D, E y F son puntos medios, entonces CD = BF = AE ♦ Coinciden todos los puntos singulares: H = I = G = O
TRIANGULO ISÓSCELES	
	♦ Solo coinciden los elementos secundarios que caen a la base. ♦ CD = Altura = Bisectriz =Transversal de Gravedad = Simetral

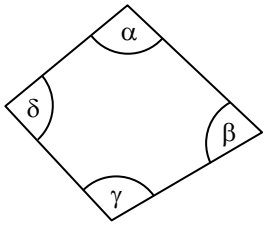
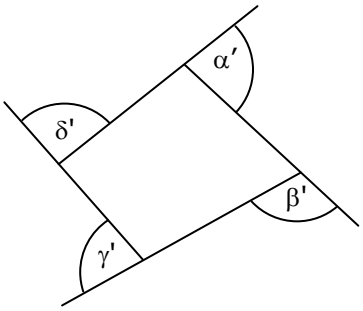
POLÍGONOS

	CLASIFICACIÓN	REPRESENTACIÓN GRÁFICA	OBSERVACIÓN
POLÍGONOS	**Cóncavo**		Al menos un ángulo interior mide más de 180°
	Convexo		Todos sus ángulos interiores miden menos de 180°

	PROPIEDADES Y CARACTERÍSTICAS
POLIGONOS CONVEXOS (n lados)	♦ Diagonales desde un vértice: $n - 3$ ♦ Cantidad total de diagonales: $\frac{n(n-3)}{2}$ ♦ Suma de los ángulos interiores: $180° (n - 2)$ ♦ Suma de los ángulos exteriores: 360°
POLIGONO REGULAR (n lados)	♦ Polígono convexo ♦ Todos sus lados son congruentes ♦ Todos sus ángulos interiores son congruentes ♦ Todos sus ángulos exteriores son congruentes ♦ Ángulo exterior: $\alpha' = \frac{360°}{n}$ ♦ Ángulo interior: $\alpha = 180° - \alpha'$

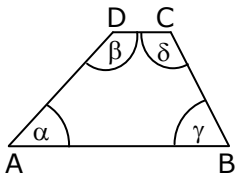
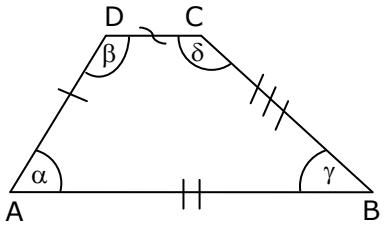
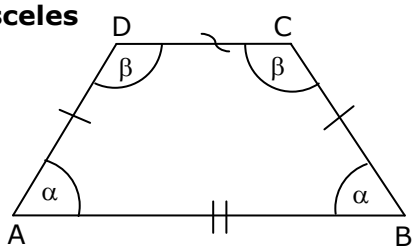
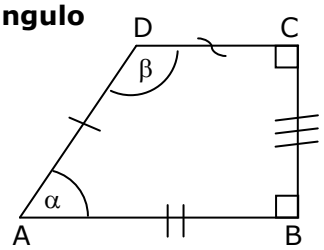
OBSERVACION	EJEMPLO
En un polígono regular, al trazar todas las diagonales desde un vértice, éstas dividen al ángulo interior en ángulos congruentes.	
El Hexágono regular es el único polígono regular que se puede subdividir en 6 triángulos equiláteros congruentes.	
Los otros polígonos regulares se subdividen en triángulos isósceles.	

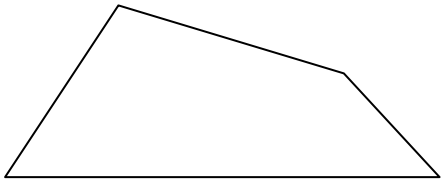
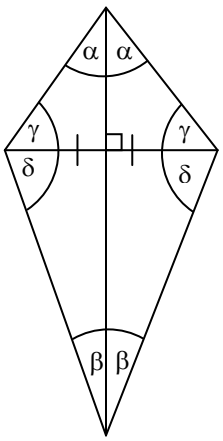
CUADRILÁTEROS (CONVEXOS)

TEOREMAS BÁSICOS DE CUADRILÁTEROS		
Suma de los ángulos interiores		$\alpha + \beta + \gamma + \delta = 360°$
Suma de los ángulos exteriores		$\alpha' + \beta' + \gamma' + \delta' = 360°$

	CLASIFICACIÓN		OBSERVACIÓN
CUADRILÁTEROS	**Paralelogramos**	♦ Cuadrado ♦ Rombo ♦ Rectángulo ♦ Romboide	Dos pares de lados paralelos
	Trapecios	♦ Escaleno ♦ Isósceles ♦ Rectángulo	Un par de lados paralelos
	Trapezoides	♦ Asimétrico ♦ Simétrico (Deltoide)	Sin lados paralelos

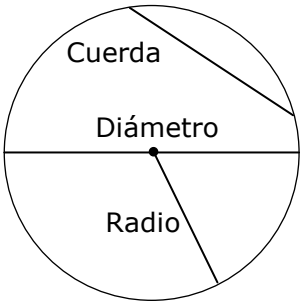
PROPIEDADES DE LOS PARALELÓGRAMOS		
PARA TODOS LOS PARALELÓGRAMOS	**CASOS PARTICULARES**	
♦ Lados opuestos congruentes ♦ Ángulos opuestos congruentes ♦ Ángulos contiguos suplementarios ♦ Las diagonales se dimidian	**Cuadrado**	♦ Diagonales perpendiculares ♦ Diagonales bisectrices ♦ Diagonales congruentes
	Rombo	♦ Diagonales perpendiculares ♦ Diagonales bisectrices
	Rectángulo	♦ Diagonales congruentes

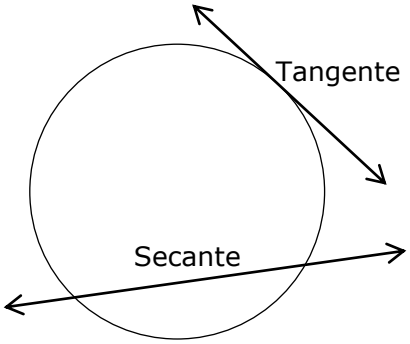
TRAPECIOS	**CLASIFICACIÓN Y REPRESENTACIÓN GRÁFICA**	**OBSERVACIONES Y PROPIEDADES**
 ($\overline{AB}$ // $\overline{DC}$) ♦ Ángulos colaterales suplementarios: $\alpha + \beta = 180°$ $\delta + \gamma = 180°$ ♦ Se llama Base a cada uno de los lados paralelos ($\overline{AB}$ // $\overline{DC}$)	**Escalenos**  $\overline{AB}$ // $\overline{DC}$	♦ Todos los lados son distintos ♦ Todos los ángulos interiores son distintos
	Isósceles  $\overline{AB}$ // $\overline{DC}$	♦ Dos pares de ángulos basales ♦ Ángulos opuestos suplementarios ♦ Diagonales congruentes
	Rectángulo  $\overline{AB}$ // $\overline{DC}$	♦ Uno de sus lados no paralelos es perpendicular a ambas bases

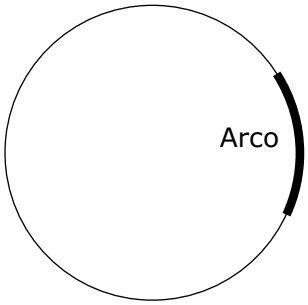
	CLASIFICACIÓN Y REPRESENTACIÓN GRÁFICA	OBSERVACIONES Y PROPIEDADES
Trapezoides	**Asimétrico** 	♦ Suma de los ángulos interiores igual a 360°
	Simétrico (Deltoide) 	♦ Diagonales diferentes ♦ Diagonales perpendiculares ♦ Una diagonal es bisectriz y simetral de la otra diagonal

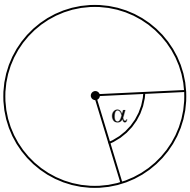
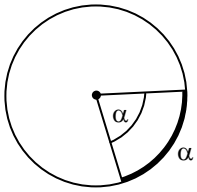
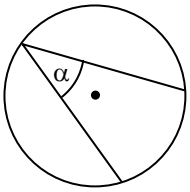
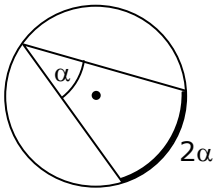
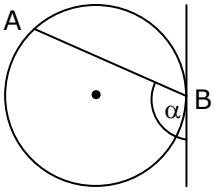
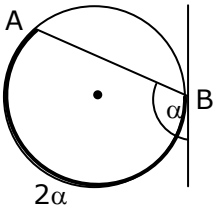
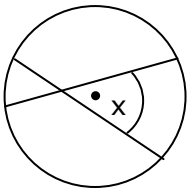
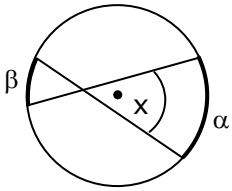
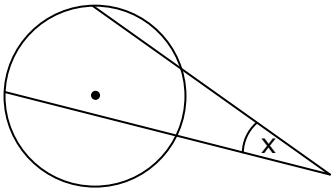
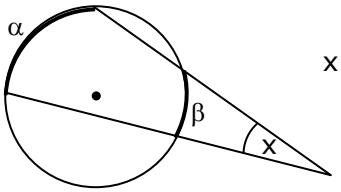
CIRCUNFERENCIA

CONCEPTOS BÁSICOS:

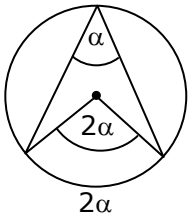
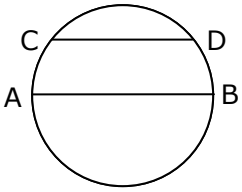
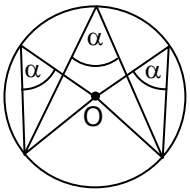
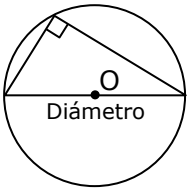
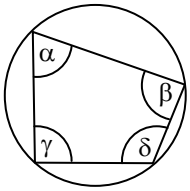
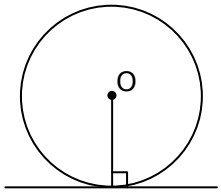


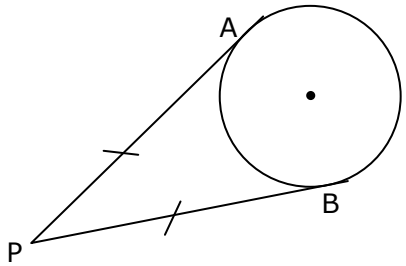
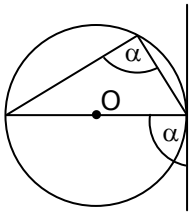
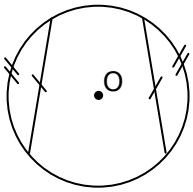
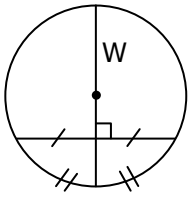
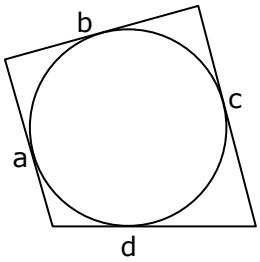




TIPO DE ÁNGULO	REPRESENTACIÓN GRÁFICA	RELACIÓN DEL ÁNGULO CON EL ARCO QUE SUBTIENDE
Ángulo del Centro		
Ángulo Inscrito		
Ángulo Semi-inscrito		
Ángulo Interior		 $x = \frac{\alpha + \beta}{2}$
Ángulo Exterior		 $x = \frac{\alpha - \beta}{2}$

TEOREMAS

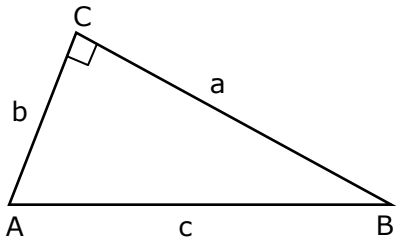
Teorema 1		Todo ángulo inscrito mide la mitad del ángulo del centro si subtienden el mismo arco.
Teorema 2		Cuerdas paralelas determinan arcos congruentes entre ellas. $\overline{AB}$ // $\overline{CD}$ ⇔ arco CA ≅ arco BD
Teorema 3		Todos los ángulos inscritos que subtienden el mismo arco son congruentes.
Teorema 4		Todo ángulo inscrito en una circunferencia que subtiende un arco de 180° es recto.
Teorema 5		En todo cuadrilátero inscrito en una circunferencia, los ángulos opuestos son suplementarios. $\alpha + \delta = 180°$ $\beta + \gamma = 180°$
Teorema 6		Un radio siempre es perpendicular a la tangente en el punto de tangencia.

Teorema 7		Si desde un punto P se trazan dos tangentes, los segmentos que unen P con los puntos de tangencia son congruentes. $\overline{PA} \cong \overline{PB}$
Teorema 8		Un ángulo inscrito mide lo mismo que un ángulo semi-inscrito si subtienden el mismo arco.
Teorema 9		Cuerdas congruentes determinan arcos congruentes. $\overline{AD} \,//\, \overline{BC} \Leftrightarrow$ arco AD $\cong$ arco BC
Teorema 10		Todo radio perpendicular a una cuerda, biseca a la cuerda y al arco correspondiente.
Teorema 11		En todo cuadrilátero circunscrito a una circunferencia, la suma de cada par de lados opuestos es la misma. $a + c = b + d$

TEOREMA DE PITÁGORAS

En todo triángulo rectángulo, el cuadrado de la longitud de la hipotenusa es igual a la suma de los cuadrados de las longitudes de los catetos.

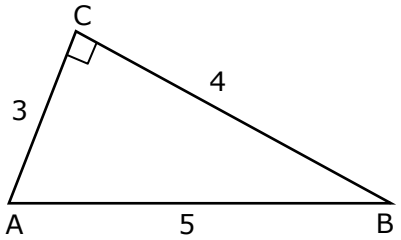
$$a^2 + b^2 = c^2$$



Tríos Pitagóricos

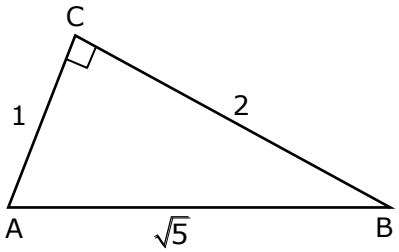
Cateto 1	Cateto 2	Hipotenusa
3	4	5
5	12	13
8	15	17

Ejemplo:



Cateto 1	Cateto 2	Hipotenusa
a	a	$a\sqrt{2}$
a	2a	$a\sqrt{5}$
a	3a	$a\sqrt{10}$
a	4a	$a\sqrt{17}$
a	ka	$a\sqrt{k^2 + 1}$

Ejemplo:

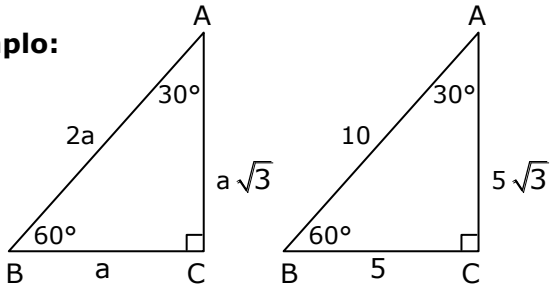


TRIÁNGULOS NOTABLES

Triángulo 30°-60°-90°

Cateto 1	Cateto 2	Hipotenusa
a	$a\sqrt{3}$	2a

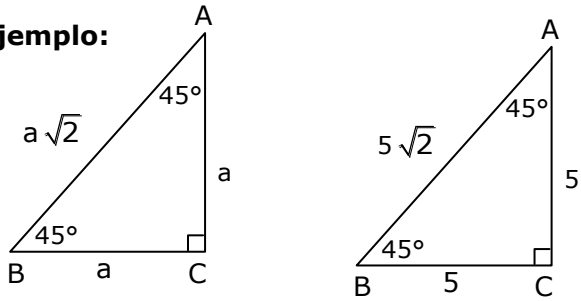
Ejemplo:



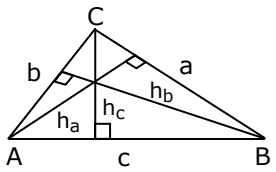
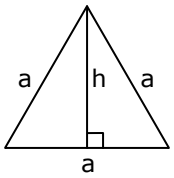
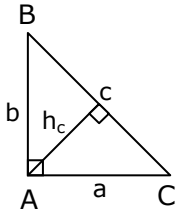
Triángulo Rectángulo Isósceles

Cateto 1	Cateto 2	Hipotenusa
a	a	$a\sqrt{2}$

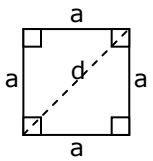
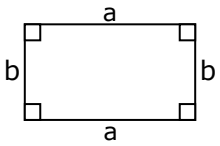
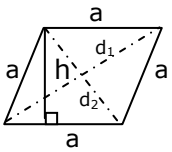
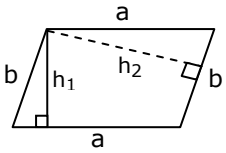
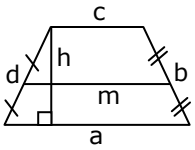
Ejemplo:



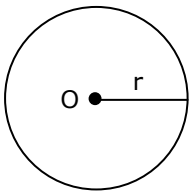
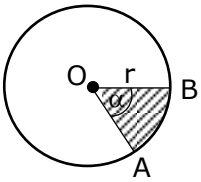
TRIÁNGULOS

ÁREAS Y PERÍMETROS			
NOMBRE	**FIGURA**	**PERÍMETRO**	**ÁREA**
Triángulo		$P = a + b + c$	$A = \frac{a \cdot h_a}{2} = \frac{b \cdot h_b}{2} = \frac{c \cdot h_c}{2}$ ♦ $A = \frac{a \cdot h_a}{2}$ ♦ $A = \frac{b \cdot h_b}{2}$ ♦ $A = \frac{c \cdot h_c}{2}$
Triángulo Equilátero		$P = 3a$	$A = \frac{a^2\sqrt{3}}{4}$ Observación: $h = \frac{a\sqrt{3}}{2}$
Triángulo Rectángulo		$P = a + b + c$	$A = \frac{ab}{2} = \frac{c \cdot h_c}{2}$ ♦ $A = \frac{ab}{2}$ ♦ $A = \frac{c \cdot h_c}{2}$

CUADRILÁTEROS

NOMBRE	FIGURA	PERÍMETRO	ÁREA
Cuadrado		♦ $P = 4a$	♦ $A = a^2$ ♦ $A = \frac{d^2}{2}$
Rectángulo		♦ $P = 2a + 2b$	♦ $A = a \cdot b$
Rombo		♦ $P = 4a$	♦ $A = h \cdot a$ ♦ $A = \frac{d_1 \cdot d_2}{2}$
Romboide		♦ $P = 2a + 2b$	♦ $A = a \cdot h_1$ ♦ $A = b \cdot h_2$
Trapecio		♦ $P = a + b + c + d$	♦ $A = \left(\frac{a + c}{2}\right) \cdot h$ ♦ $A = m \cdot h$ donde $m = \frac{a + c}{2}$

CIRCUNFERENCIA Y CÍRCULO

NOMBRE	FIGURA	PERÍMETRO	ÁREA
Circunferencia y Círculo		♦ $P = D \cdot \pi$ ♦ $P = 2 \cdot \pi \cdot r$ D: diámetro	♦ $A = \pi \cdot r^2$
Sector Circular		♦ P = Arco AB + 2r Donde: Arco AB $= \left(\frac{\alpha}{360^o}\right) \cdot 2 \cdot \pi \cdot r$	♦ $A = \left(\frac{\alpha}{360^o}\right) \cdot \pi \cdot r^2$